

Chapter 03

Cost of Direct Materials, Cost of Direct Labour and Cost of
Overhead Calculation

Exercise

Arden Company reported the following costs and expenses for the most recent month:

Direct materials	\$80,000
Direct labor	\$42,000
Manufacturing overhead	\$19,000
Selling expenses	\$22,000
Administrative expenses	\$35,000

Required:

1. What is the total amount of product costs?
2. What is the total amount of period costs?
3. What is the total amount of conversion costs?
4. What is the total amount of prime costs?

Costing for Direct Materials

1. FIFO (First-In, First-Out) Method

Common when material prices are rising; reflects the actual flow of goods.

Advantage: Provides realistic valuation of closing stock.

Limitation: Complicated when many purchase lots exist.

2. LIFO (Last-In, First-Out) Method

Sometimes used in industries where price rises rapidly.

Advantage: Matches current cost against revenue.

Limitation: Not allowed under some accounting standards; older stock may remain unsold.

3. Weighted Average Cost Method

Advantage: Smooths out price fluctuations; simple to apply.

Limitation: May not reflect actual flow of materials.

ABC Ltd manufactures electronic gadgets. During the month of August 2025, the company purchased and issued a particular raw material “X” as follows:

Date	Transaction	Quantity (units)	Unit Cost (\$)
Aug 1	Opening Stock	100	50
Aug 5	Purchase	200	55
Aug 12	Purchase	150	60
Aug 15	Issue to Production	250	?
Aug 20	Purchase	100	65
Aug 25	Issue to Production	200	?

Calculate the cost of materials issued for production on Aug 15 and Aug 25, and the closing stock at the end of Aug, using:

1. FIFO (First-In, First-Out)
2. LIFO (Last-In, First-Out), and
3. Weighted Average Cost Method

Costing for Direct Materials

4. Specific Identification (Actual Cost Method)

Each material item is tracked and issued at its actual purchase cost.

Use: High-value or unique items (e.g., machinery, jewelry).

Advantage: Exact cost of materials is recorded.

Limitation: Impractical for bulk or low-cost items.

5. Replacement/Current Cost Method

Materials are valued at current market or replacement cost instead of historical cost.

Use: Useful for inflationary periods.

Advantage: Reflects current cost of production.

Limitation: Can fluctuate frequently; not suitable for long-term valuation.

Precision Machining Co.'s high-grade titanium alloy is a specialized product where each batch can be uniquely identified. The company's internal management accounting system is evaluating alternative costing methods for strategic decision-making. The transactions for October are detailed below, including unique batch numbers:

Date	Transaction	Batch No.	Units (kg)	Cost per kg (£)	Total Cost (£)
Oct. 1	Beginning Inventory	T-101	400	50.00	20,000
Oct. 5	Purchase	T-215	600	55.00	33,000
Oct. 12	Issue to Job A101		700	?	?
Oct. 18	Purchase	T-308	500	60.00	30,000
Oct. 25	Issue to Job B202		400	?	?
Oct. 29	Purchase	T-419	300	62.00	18,600

Additional Information:

The production manager's reports show that the issue on October 12 for Job A101 specifically used:

- 200 kg from beginning inventory batch T-101
- 500 kg from the October 5 purchase batch T-215

The issue on October 25 for Job B202 specifically used:

- 200 kg from the October 18 purchase batch T-308
- 200 kg from the remaining beginning inventory batch T-101

On October 25, the replacement cost (market price) to purchase more titanium alloy was £61.50 per kg.

Required:

Calculate the total direct material cost charged to Job A101 and Job B202, and the value of the ending inventory on October 31st using the following two methods:

a) Specific Identification Method

b) Current Cost (Replacement Cost) Method (Use the replacement cost given for the issue on Oct. 25. For the Oct. 12 issue, assume the replacement cost on that date was £56.50 per kg).

Costing for Direct Labour

- Direct Labour is the cost of that labour which is expended in altering the construction, composition or condition of the product.
- Indirect Labour cost is the wages paid to those workers who are not directly engaged in converting raw materials into finished products.

Costing for Direct Labour

1. Time-Based Systems (Day Work)

$\text{Wages} = \text{Time worked} \times \text{Rate per hour}$

2. Piece-Rate Systems

$\text{Wages} = \text{Units produced} \times \text{Rate per unit}$

3. Premium Bonus Plans (Incentive Schemes)

a. Halsey Plan

b. Rowan Plan

c. Taylor's Differential Piece Rate System

a. Halsey Plan

It is a simple combination of time and piece method of wage payment. Under this plan, amount of bonus depends upon the **time saved by the worker**. A standard time is fixed for each job and if a worker completes the job in less than the standard time, he gets wages for the time worked plus a bonus equal to 50% of the value of time saved.

Total Earnings = (Time taken x Rate Per Hour) + 50%(Time Saved x Rate Per Hour)

Time saved = Standard time – Time Taken

b. Rowan Plan

This plan was introduced by David Rowan and is similar to Halsey plan, except in the calculation of the amount of bonus. In this plan, bonus is that proportion of the wages of the time taken which the time saved bears to the standard time. Its formula is:

Bonus = Time saved / Standard time x Time taken x Rate per hour

Total Earnings = (Time taken x Rate per hour) + Bonus.

Halsey Plan and Rowan Plan - Exercise

ZED Limited is working by employing 50 skilled workers. It is considered the introduction of incentive scheme-either **Halsey Scheme (with 50% bonus)** or Rowan scheme of wage payment for increasing the labour productivity to cope up the increasing demand for the **product by 40%**. It is believed that proposed incentive scheme could bring **about an average 20%** increase over the present earnings of the workers; if could act as sufficient incentive for them to produce more. Because of assurance, the increase in productivity has been observed as revealed by the figures for the month of April, 2004.

- Hourly rate of wages (guaranteed) Rs. 30
- Average time for producing one unit by one worker at the previous
- Performance (This may be taken as time allowed) 1.975 hours
- Number of working days in the month 24
- Number of working hours per day of each worker 8
- Actual production during the month 6,120 units

Required: Calculate the effective rate of earnings under the Halsey scheme and the Rowan scheme.

c. Taylor's Differential Piece Rate System

- F.W. Taylor as a part of the scheme of scientific management. Under this system, the standard task is established after careful time and motion study and two piece rates are set. The main features of the scheme may be given as below:
- Day wages are not guaranteed
- A standard time for jobs is established
- Two piece rates are fixed. If the worker does the work within the standard time, he receives the higher piece rate, whereas if he takes longer time he receives the lower piece rate.
- Usually these rates are 83% of the piece work rate for inefficient workers and 175% of the piece rate for efficient workers.

Taylor's Differential Piece Rate System

A company sets the standard output of a worker at 200 units per day. Taylor's Differential Piece Rate System is applied with the following rates:

Lower piece rate: \$2.00 per unit (if output < 200 units)

Higher piece rate: \$3.00 per unit (if output $\geq$ 200 units)

During one week (5 working days), three workers produce the following outputs:

Worker	Day 1	Day 2	Day 3	Day 4	Day 5
A	180	190	200	210	220
B	150	160	170	180	190
C	220	230	210	205	215

- Calculate the daily wages of each worker using Taylor's Differential Piece Rate System.
- Find the total weekly earnings of each worker.
- Compare the average earnings per day of Worker B and Worker C.

Costing for Overheads

1. Traditional / Absorption Costing Methods
2. Activity-Based Costing (ABC)
3. Direct Allocation Method (Primary Distribution)
4. Machine Hour Rate Method
5. Labour Hour Rate Method
6. Unit of Output Method

Traditional / Absorption Costing Methods

1. Collect all indirect costs (overheads)
2. Classify by department (production/service)
3. Apportion common overheads to departments
4. Reapportion service department costs to production departments
5. Select absorption base (DLH, MH, etc.)
6. Absorb overheads to products using OAR
7. Adjust for under- or over-absorption

A company manufactures filing cabinets. The company applies all manufacturing overhead to products based on direct labor cost. Selling and administrative expenses are treated as period costs. The company's total budgeted costs and activity for the year are as follows:

Data:

- Total Manufacturing Overhead: \$500,000
- Total Selling and Administrative Overhead: \$300,000
- Total units produced: 1,000 units
- Direct labor cost per unit: \$90

The company received an order from a customer, OfficeMart, with the following details:

- Total number of filing cabinets ordered: 80 units
- Selling price per unit: \$795
- Direct materials per unit: \$185
- Direct labor per unit: \$90

Required: Using the absorption costing method, calculate the gross margin for the OfficeMart order.

Activity-Based Costing (ABC)

1. Identify activities (e.g., machine setups, quality checks, material handling).
2. Collect costs for each activity (cost pool).
3. Identify cost drivers (e.g., number of setups, inspection hours).
4. Absorb costs to products based on actual activity usage.

Activity-Based Costing (ABC)

Let's assume that OfficeMart places different orders next year, purchasing higher-end filing cabinets more frequently, but in smaller quantities per order. Enter the following data into your worksheet

Data				
Manufacturing overhead	\$500,000			
Selling and administrative overhead ...	\$300,000			
	Assembling Units	Processing Orders	Supporting Customers	Other
Manufacturing overhead	50%	35%	5%	10%
Selling and administrative overhead	10%	45%	25%	20%
Total activity	1,000 units	250 orders	100 customers	
OfficeMart orders:				
Customers	1 customer			
Orders	20 orders			
Total number of filing cabinets ordered ...	80 units			
Selling price	\$795			
Direct materials	\$185			
Direct labor	\$90			

a. What is the customer margin under activity-based costing?

Direct Allocation Method (Primary Distribution)



A factory has 2 Production Departments (Soaking and Liming) and 2 Service Departments (HR and Accounting).

Total overheads to distribute:

Rent = Tk 60,000

Power = Tk 30,000

Supervisor Salary = Tk 20,000

Department	Floor Area (sq. ft.)	HP of Machines	No. of Workers
Soaking	2,000	50	25
Liming	1,000	30	15
HR	500	20	10
Accounting	500	0	5
Total	4,000	100	55

Requirement: Now allocate the overhead costs using direct allocation method (primary).

Machine Hour Rate Method

- Overheads absorbed based on machine hours.
- Suitable for machine-intensive industries.

$$\text{Machine Hour Rate} = \frac{\text{Total Factory Overheads}}{\text{Total Machine Hours}}$$

Labour Hour Rate Method

- Overheads absorbed based on direct labour hours worked.
- Suitable for labour-intensive industries.

$$\text{Labour Hour Rate} = \frac{\text{Total Factory Overheads}}{\text{Total Labour Hours}}$$

Unit of Output Method

- Overhead absorbed per unit of production.
- Simple but works only if products are homogeneous.

$$\text{Overhead Rate per Unit} = \frac{\text{Total Units Produced}}{\text{Total Factory Overheads}}$$